



Society of Petroleum Engineers

SPE-229828-MS

Advanced Techniques for Subsea Umbilical Pre-Commissioning in Deepwater Tiebacks - Concept and Application from the Begonia Development, Offshore Angola

Giulio Donolo, Fano, Italy, EnerMech

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229828-MS](https://doi.org/10.2118/229828-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

As offshore developments are increasingly located in deepwater and ultra-deepwater environments, the integration of new infrastructure into existing subsea systems becomes more complex.

Such developments typically involve the addition of new wells, flowlines and umbilicals which present unique challenges from an installation and pre-commissioning point of view, particularly for post installation testing of subsea umbilical systems.

Subsea umbilicals are critical conduits for hydraulic fluids, production chemicals and communication signals, and must be verified for integrity before being commissioned.

In brownfield expansions, where umbilicals are terminated subsea at both ends, conventional surface-based post installation testing is no longer viable. This necessitates alternative approaches that can operate effectively in challenging subsea environments ranging from 900 to over 2,200 meters.

Two field-proven pre-commissioning technologies have been developed to address this gap. The first involves the deployment of a ROV-powered, self-contained subsea test pump (USTP), suitable for testing systems with moderate volume-to-pressurize (VTP) requirements. The second employs twin single-section, high-pressure composite downlines deployed from a topside vessel to deliver test fluids at depth, suitable for high-volume test applications.

This paper presents the technical framework, operational parameters, and deployment considerations of such solutions with a particular focus on the EnerMech in-house developed USTP. It also details the USTP application in the Begonia development, offshore Angola, where an umbilical subsea network of production and water injection umbilicals were tested in situ as a unique and entire system.

The Begonia development case study outlines system configuration, fluid management, test sequencing, data acquisition, and the integration of field personnel and logistics across regional teams.

The deployment of advanced umbilical testing systems demonstrated reliable performance, validated umbilical integrity, and contributed to the successful tieback of the Begonia subsea wells to the existing Pazflor production system, a vast subsea gathering network, the most complex ever built in Angola: 180 kilometres of lines tying in 49 subsea wells, 10,000 metric tons of subsea equipment and the giant Pazflor floating production, storage and offloading (FPSO) vessel.

Introduction

The requirement for reliable pre-commissioning of subsea umbilicals in deepwater environments is increasing as more brownfield tiebacks are executed globally. In such projects, surface access to umbilical terminations after installation completion are unavailable, rendering conventional hydraulic and electrical testing methods infeasible. In addition, increased operational complexity, driven by deeper water, longer tieback distances, and stricter cleanliness and pressure requirements, has elevated the need for innovative testing solutions.

This paper introduces two complementary subsea umbilical testing methodologies developed to meet these needs. Each is optimized for specific project scenarios and fluid volumes. Following a technical overview of each system, a field case study from the Begonia development, offshore Angola demonstrates the methodologies in action, highlighting both technical and logistical outcomes.

Statement of theory and definitions

Subsea umbilicals

Subsea umbilicals are integrated bundles containing hydraulic control lines, chemical injection lines (e.g., mono-ethylene glycol [MEG], corrosion inhibitors), electrical power conductors and optical communication fibers. Their function is to enable operation, monitoring and chemical treatment of subsea production infrastructure.

Each umbilical component requires specific testing and monitoring during the various phases of the installation process. While the majority of these tests are performed from the installation vessel when at least one termination is accessible, the final test (Post Installation Test) upon installation completion requires the testing equipment to be tied-in directly subsea at one of the two terminations.

The most common failure modes observed in subsea umbilicals during inspection and testing campaigns include:

- Micro-leaks in hydraulic conduits;
- Internal corrosion driven by fluid contamination;
- Signal attenuation in optical fibers due to water ingress;
- And degradation of electrical insulation under high hydrostatic pressure.

These failure modes require testing procedures specifically designed to detect anomalies at an early stage, as undetected issues could compromise the long-term reliability of the entire production system.

The selection of testing methodologies must also align with applicable international standards. In particular, API 17E and ISO 13628-5 define the requirements for design, manufacturing, and testing of umbilicals, while ISO 4406 and ISO 13628 establish cleanliness criteria for hydraulic fluids and acceptance thresholds during pressure testing. Integrating these best practices into the pre-commissioning phase ensures methodological consistency and traceability of results, providing conformity with industrial audit requirements.

The integrity verification of the hydraulic/chemical lines, the conductors and Fiber Optic components is mandatory. For the hydraulic testing, a key factor in selecting the appropriate testing method is the volume required to pressurize the hydraulic cores to the required test pressure (VTP). The VTP determines whether an ROV-deployed system or topside-supported downlines are suitable.

Standard tests include:

Hydrostatic testing: Verifies pressure containment integrity of fluid cores. Fluids used for testing shall meet ISO cleanliness standards. Cleanliness shall be verified in field prior to the testing.

Electrical and Fiber Optic testing to ensure electrical/FO circuit continuity (CR), and insulation (IR). TDR or OTDR is used where fault finding is required.

Description and application of equipment and processes

ROV-powered subsea test pump (USTP)

The USTP system is a modular, self-contained test skid deployed to the seabed. It is powered via a work-class ROV and features:

- Independent test loops: Two fluid systems operate in parallel, typically for hydraulic and MEG-based fluids. Each system includes a dedicated pump and fluid reservoir.
- Test pumps: High-pressure, self-priming pumps designed to efficiently convert ROV hydraulic power into high-pressure output. Available in various models to suit the pressure and flow rate requirements of specific testing operations. These pumps are completely free of rotating parts, and their compact size, lightweight design and robust construction make them extremely versatile and reliable. They can be easily installed, removed and swapped within the USTP frame to meet specific project needs.
- ROV integration: ROV delivers hydraulic power via a custom hot stab connection. The rate of pressurization is adjusted by modulating ROV hydraulic flow.
- Fluid reservoirs: Two integrated reservoirs, each pressure-compensated, with storage capacity of ~170 liters. Additional capacity can be added via subsea chemical baskets if required.
- Fluid cleanliness: All fluids are filtered and tested to meet ISO cleanliness standards prior to reservoir filling and subsea deployment. Line flushing and sampling confirm compliance.
- Data monitoring: Pressures are monitored in real-time using subsea data loggers and Keller gauges. In addition, an optical uplink transmits pressure data topside via the ROV communication channels for immediate assessment.
- Depressurization control: Upon test completion, depressurization is controlled using a calibrated flow-reduction device sized to the specific umbilical VTP and abandonment pressure requirements.

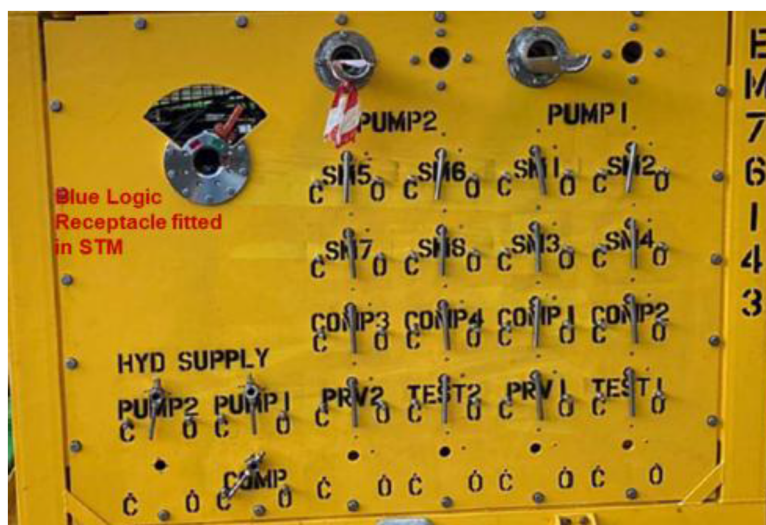


Figure 1—USTP ROV Front Panel



Figure 2—USTP integrated fluid reservoirs

This approach is effective for short to medium-length umbilicals or networks of umbilicals, particularly where testing with multiple fluid types is required. Additionally, the system can remain subsea between different operations and can be used even in weather conditions that would prevent the use of conventional testing methods such as downlines.

Twin single-section composite downlines

For high VTP scenarios, a surface-pumped system utilizing twin composite downlines was developed. Key features include:

- Downline configuration: Each line is 1,500 meters long with a 0.75" inner diameter. The lines are single-section and self-supporting.
- Material properties: Composite construction with smooth bore prevents fluid contamination. Rated to 10,000 psi (690 barg) MAWP.
- Collapse resistance: External collapse resistance of 100 bar supports deployment to ~1,000 meters, with optional safety systems for deeper depths.
- Topsides equipment: Pumps and fluid reservoirs are located on the support vessel,

Deployment and recovery are typically completed within an hour at water depths up to 1,350 meters. This technique is favored for large umbilicals or when extended test durations are required.

The USTP represents not only a controlled pressurization solution but also an engineering evolution in terms of modularity and adaptability. The materials selected for the frame and reservoir construction are designed to withstand repeated immersion and recovery cycles. The system's modularity enables configuration for different operational scenarios allowing for example the addition or removal of auxiliary reservoirs and the installation and fit up of different test pump models as required by different projects. This makes the unit highly scalable and reusable across multiple projects, thereby optimizing overall campaign costs. Regarding the twin composite downlines, the choice of composite material over conventional metallic pipes provides tangible advantages. Beyond the significant reduction in linear weight, which simplifies deployment and retrieval operations, the smooth-bore internal surface limits fluid cross-contamination and reduces pressure drop. Compared with metallic pipes, composite downlines eliminate the risk of cyclic fatigue collapse and offer higher intrinsic resistance to lateral marine currents. As a result, the overall safety of operations is enhanced, and vessel time required for testing is reduced, directly lowering the daily cost of offshore campaigns.

USTP case study presentation of data and results

Project overview: Begonia Development, Offshore Angola

The Begonia field is located offshore Angola within the block B17/06, and adjacent to the B17 FPSOs of Girassol, Dalia, Pazflor and CLOV. The Begonia development targets an upper-Miocene reservoir, collected through the Pazflor subsea production system via a subsea-to-subsea tie-back. It includes five firm wells: three production and two water injection.

The three production wells are gathered through a multiphase production flowline. The two water injection wells are connected through an extension of the existing Pazflor water injection system. Production wells and water injection wells are supported by an extension of Pazflor's subsea distribution network for power/communication, hydraulic and chemicals.

The Pazflor subsea production system is connected to the spread moored FPSO, 30 km south of the Begonia field and 150km west from the coast, via a series of subsea production and injection lines, umbilicals and associated risers.

- **Location:** Block B17/06
- **Water depth:** 400–775 meters
- **Tieback system:** Subsea-to-subsea connection to the existing Pazflor development's production infrastructure
- **Execution window:** February – July 2025 (5 months)
- **Scope focus:** Subsea Umbilical Network Testing (Production and Water Injection)Z

Scope of work: subsea umbilical network testing

The network testing campaign focused on the production and water injection umbilicals across water depths ranging from 550 to 750 meters. The umbilicals, ranging from 2 km to 9 km in length, included two production and three water injection umbilicals. The test scope included a Network Testing of the entire umbilical system, CR & IR Electrical Testing, and Pressure Testing of the Hydraulic & Chemical Cores, all carried out entirely subsea. The electrical testing has been performed from subsea by using third-party testing tools while Hydraulic Testing has been performed via the USTP herein described.

In addition to the challenges of conducting pressure testing entirely subsea in deep water, the network testing required larger fluid volumes than the USTP could currently hold.

These tests were conducted without recovering any components to the surface, requiring precise subsea fluid management and control.

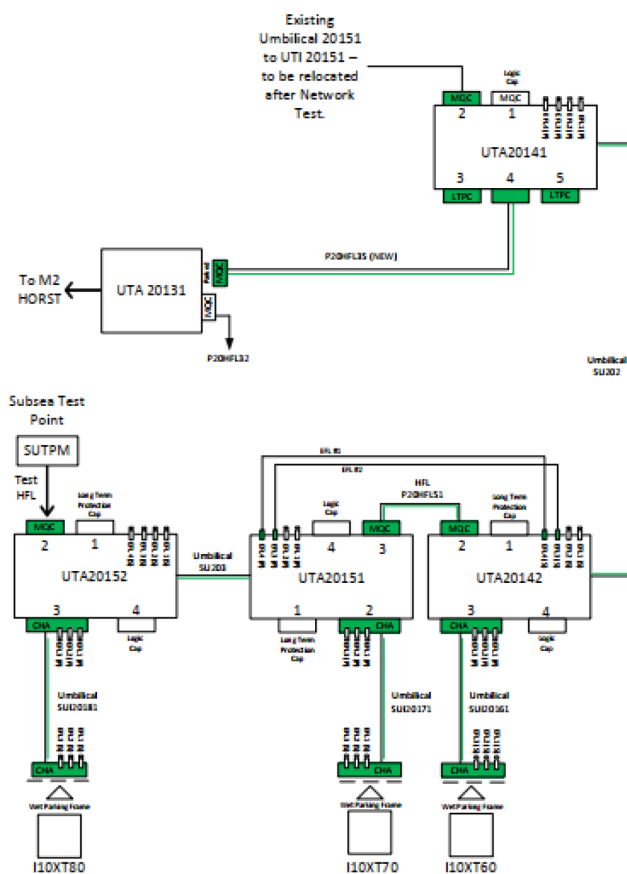


Figure 3—Subsea umbilicals network

System deployment

The umbilical pre-commissioning activities were split into two phases. The first one aimed to test the network of two main production umbilicals for a total length of approx. 15 km. Tests consisted of electrical and hydraulic pressure tests to confirm the integrity of the two main production umbilicals. Both main umbilicals terminated into UTAs (Umbilical Termination Assemblies) and were connected together via hydraulic and electrical flying leads

The second aimed to test the entire umbilical network including three water injection umbilicals for a total length of approx. 22.5 km.

Hydraulic and chemical lines were pressure tested by the deployment of a subsea SUTPM. The Testing system was set up by an ROV connecting the Test HFL (Hydraulic Flying Lead) to the Umbilical Termination on one side, which provided access to the fluid line groups within the umbilical and allowed them to be tested separately. The other side of the Test HFL split into 4 individual lines with an SQC (Single Quick Connect) on each termination – see Fig. below – which were connected, to the USTP receptacles.

Connection of the SQCs to the USTP enabled each of the umbilicals lines or group of lines to be pressure tested with respect to maximum test pressure and fluid type as required by the test procedure.

As an example, the Production umbilicals groups were split as follows:

Group 1: HP Control – Fluid HW433 – Test Pressure 690 bar

Group 2: LP Control – Fluid HW433 – Test Pressure 690 bar

Group 3: Chemical/Service – Fluid MEG 50/50 – Test Pressure 759 bar

Group 4: MeOH injection – Fluid MEG 50/50 – Test Pressure 380 bar

The USTP and the chemical basket were lowered to the seabed by the Pre Commissioning Support Vessel crane and positioned at a suitable location near the Umbilical Termination Assembly (UTA) identified as the injection point for the umbilical system test. The subsea HFL Deployment basket containing the Test

HFL (Hydraulic Flying Lead) was lowered by the Pre Commissioning Support Vessel crane to the seabed and located in between the USTP and the umbilical UTA to allow easy access for the Test

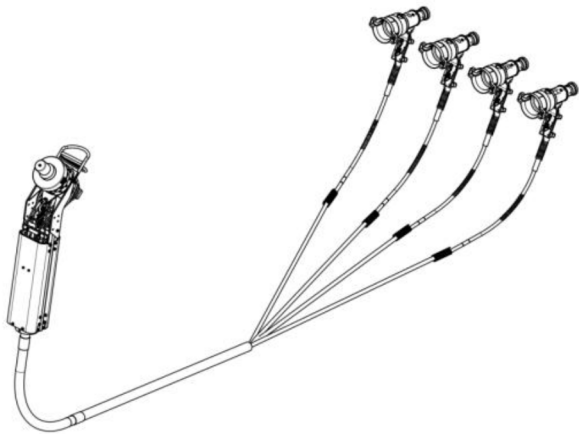


Figure 4—Hydraulic flying lead

Since the HFL Test was approximately 20 metres long, the HFL deployment basket was positioned at approx. 3 to 4 metres from the UTA, while the USTP was positioned at approximately 10 to 12 metres from the UTA.

The Chemical Basket was instead positioned in close proximity of the USTP (at approx. 3m) in order to minimize the pressure losses of the test fluids during the suction phase by the test pump.

The USTP connected each of its pressurization circuits to the Chemical Basket by use of two low pressure hoses with single hotstab connectors

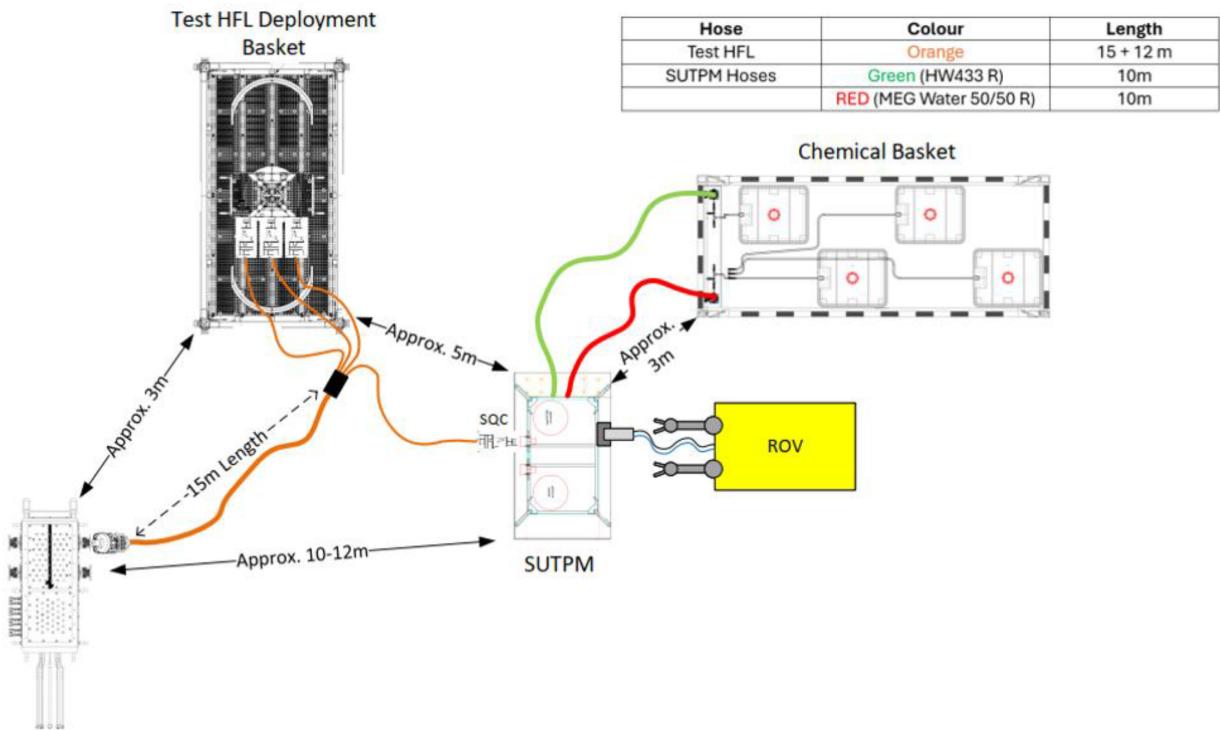


Figure 5—Subsea temporary test assets schematic and connections

The ROV connected its hydraulic work-pack to the USTP via a 3-way hydraulic stab and the USTP pumps were ‘driven’ by the hydraulic fluid circulated from the ROV.

The USTP provides two high pressure outlets which are connected to the required Testing lead Tail based on the group of cores to be tested. Within the USTP, each pressurization circuit is fitted with PRV's that allow test to be conducted safely by protecting the umbilicals’ group of cores from any over-pressurization.

During operations the test pressures and temperature are visible by ROV cameras and logged by a Keller datalogger connected to each of the USTP output lines. Pressure is also monitored on a laptop display in the ROV control room since the pressure data are transmitted via the optical link technology through the ROV control umbilical,

Chemical volume management:

Due to fluid volume demands exceeding the onboard capacity of the USTP, EnerMech deployed an auxiliary Subsea Chemical Basket. This allowed the required volumes of MEG/water and hydraulic fluids to be delivered to the umbilical cores during the staged pressure testing process. The table below reports the required fluid volumes to perform both tests



Figure 6—Subsea Chemical Basket

Table 1—Chemical volume management

Test	HW443 R	MEG/Water 50/50 R
Network	313.2	1568.4
Backbone	211.2	1538.1
Total (Litres)	524.4	3106.5

Pump configuration:

A dedicated low-flow pressurizing pump was selected and integrated into the USTP, optimized for the specific flowrate and pressure requirements of the umbilical test program.

Each test was performed at a flow rate ranging between 10 to 20 lt/min to achieve a pressurization rate in a range of 5 - 10 bar/min, which has led to a total pressurization time of not more than 2 hr for each test.

Data collection:

Real-time pressure and test status were monitored through the optical uplink system and recorded via redundant subsea data loggers for post-test verification.

Execution strategy

- **Engineering coordination:** the integration of onshore and offshore teams: the operational team in Angola worked in constant coordination with the engineering center in the UK, reducing response times to technical contingencies. Real-time communication systems enabled the transfer of data and interim results directly to shore, minimizing delays and facilitating informed decision-making.
- **Subsea umbilical test pump & manifold (SUTPM):**
Function tested and set up in EnerMech's Angola facility and used as the primary subsea pressurization unit.
- **Workforce structure:**
A four-person subsea team, comprising engineers, technicians, and supervisors was assigned to the field project execution supported by a core office team. Personnel continuity was maintained throughout both the preparatory and operational phases.

Performance results

- All umbilicals tested - spanning both production and water injection systems - successfully passed CR, IR, and pressure tests under the defined acceptance criteria.
- The system achieved pressure holds at 399, 725, and 797 barg for the specified durations, confirming the integrity of entire newly installed subsea umbilical network.
- Fluids utilized met required ISO cleanliness standards following flushing and sampling procedures before deployment.
- The subsea chemical basket enabled seamless fluid changeover and replenishment without recovering subsea assets. Efficient integration between EnerMech and client field teams contributed to maintaining the planned project timeline.

Operational impact

The execution of deepwater subsea network testing using integrated tools and systems demonstrated the viability of entirely subsea-based umbilical validation, a key requirement in modern deepwater field developments where surface terminations are not accessible.

EnerMech's ability to adapt USTP infrastructure, augment subsea fluid capacity, and execute pressure tests at near-800 barg pressures safely and without operational delay underscored the technical robustness of the approach.

The data collected during the campaign demonstrated high stability of the test pressures, with minimal fluctuations during hold periods. The ability to carry out all activities entirely subsea reduced the number of recovery and redeployment operations, lowering the risks typically associated with deck handling. These improvements translated into increased operational efficiency and a significant reduction in overall campaign costs.

Feedback from the project partner highlighted both technical competence and professionalism, contributing to the establishment of a strong working relationship for future subsea campaigns.

Conclusions

Two distinct methodologies for subsea umbilical testing have been successfully developed to brownfield expansions in deepwater environments. The ROV-powered test pump provides a controlled and self-contained option for smaller umbilicals, while the twin composite downlines offer an effective solution for higher-volume test requirements.

Application in the Begonia development confirmed the operational viability of the ROV-powered test pump, achieving full system validation within a defined execution window. These approaches allow for flexibility, meet international testing standards and are adaptable to regional deployment.

As subsea infrastructure continues to expand into more remote and deeper environments, the ability to validate umbilical systems reliably and efficiently will remain critical. The deployment of tailored subsea testing systems, supported by trained personnel and integrated project teams, offers a proven strategy to meet this demand.

The experience gained in the Begonia project confirms that these technologies are transferable to a wide range of offshore scenarios, including ultra-deepwater environments and even more extensive umbilical networks.